

The fork_wall Environment:

A Sequential Experimental Study of a Recurrent Agent’s Beliefs, Skills and Decisions in Partially Observed Navigation

Abstract

This chapter develops, end to end, the experimental testbed used throughout the thesis: the **bridge_tunnel fork_wall** task, a partially observed navigation problem in which the map’s hidden *category* must be inferred from terrain evidence and read out in a discrete door choice. We proceed in seven steps: (§1) the environment; (§2) the procedural map dataset; (§3) training recurrent PPO agents with and without an auxiliary belief loss, separating task *completion* from *decision quality*; (§4) behaviour, including generalisation to out-of-distribution maps and the three crossing regimes (build / mine / avoid); (§5) what the recurrent state encodes, via linear probes; (§6) belief formation along trajectories; and (§7) whether the agent’s state contains its *future* crossing strategy before the obstacle is reached — a minimal test of model-free planning.

1 The environment

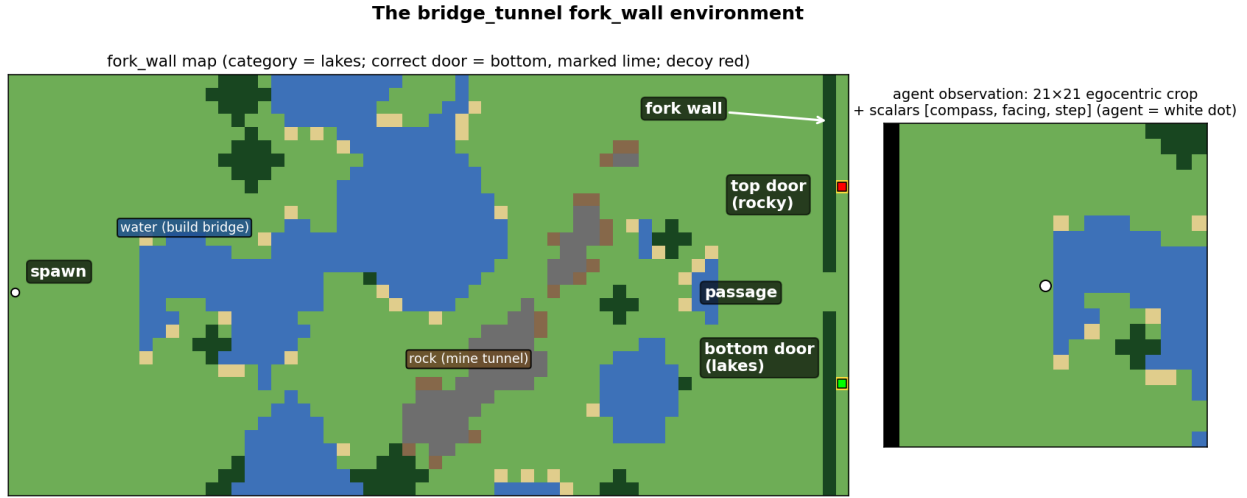


Figure 1: The fork_wall task. *Left:* a full map. The agent spawns at the west edge (white dot) and must cross a procedurally generated terrain field — water it can bridge (*build*), rock it can tunnel through (*mine*), impassable trees, and open grass — to reach a wall at the east edge. The wall has a single passage; behind it lie *two* doors. Only the door matching the map’s hidden category pays reward: bottom for **lakes**, top for **rocky**, either for **balanced**. *Right:* the agent’s observation — a 21x21 egocentric crop of tile identities plus scalar features (goal compass, facing, step count). The category is never observed directly: it must be *inferred* from the terrain statistics seen en route.

Why this design. The task couples three computations that we want to study separately in one agent: (i) *navigation* with terrain-dependent skills (bridging water costs a build action per cell; mining rock likewise); (ii) *belief formation* — the map category is a latent variable whose evidence is

spatially distributed and noisy; and (iii) a *discrete decision* (the door) whose only correct predictor is that belief. Because the doors are physically identical and spatially separated from the evidence, door accuracy is a pure behavioural readout of the agent’s belief.

Mechanics. Actions: four movements, **build** (place a bridge tile on adjacent water) and **mine** (convert adjacent rock to walkable ground). Reward: a per-step slack penalty, potential-based shaping toward the correct door, and a terminal bonus for the category-correct door. Episodes are capped generously (800 steps) so that timeouts indicate genuine failure to cross rather than tight scheduling.

2 The map dataset

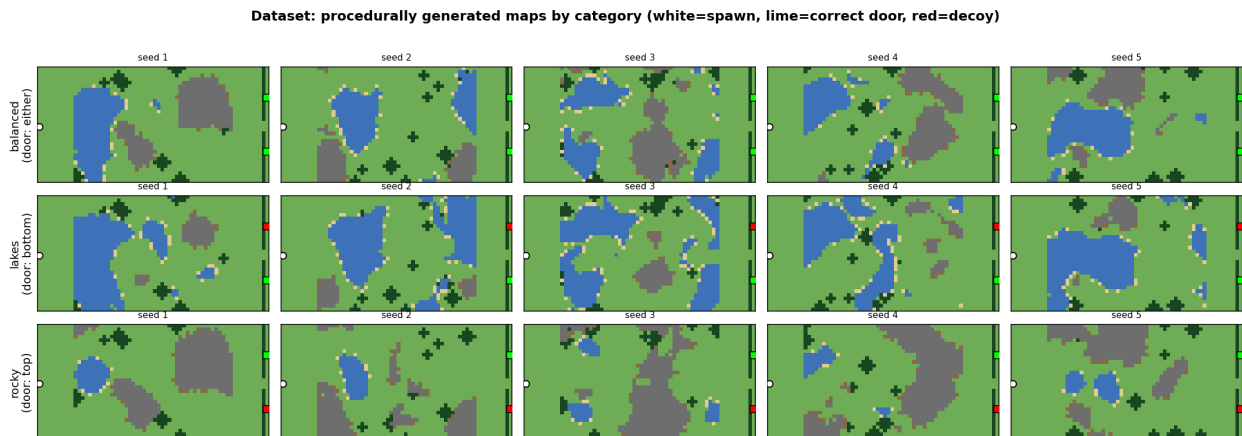


Figure 2: Procedural maps by category. Rows: **balanced**, **lakes**, **rocky**; columns: generator seeds. Categories share one generator and differ only in the water/rock coverage fractions, so no single tile identifies the category — classification requires integrating terrain statistics. Training samples maps procedurally (fresh seed per episode); evaluation uses held-out seeds ($\geq 10\,000$).

The three categories are deliberately *overlapping* distributions: **balanced** maps contain both lakes and rock formations, and individual local views of a **lakes** map can look rocky. The category is thus a genuinely latent, spatially extended variable — the property that makes the belief analyses of §5–7 meaningful.

3 Training: auxiliary belief loss vs. none

We train recurrent PPO (GRU core, 128 units; identical architecture and hyperparameters, seed 2, 6M environment steps) in two variants: with an **auxiliary belief loss** (a linear head on the GRU state classifying the map category, coefficient 0.3, gradients flowing into the recurrent core) and with **no auxiliary loss**. The auxiliary head never affects the reward; it only shapes the representation.

fork_wall PPO+GRU training: auxiliary belief loss (2 seeds) vs none (identical hyperparameters)

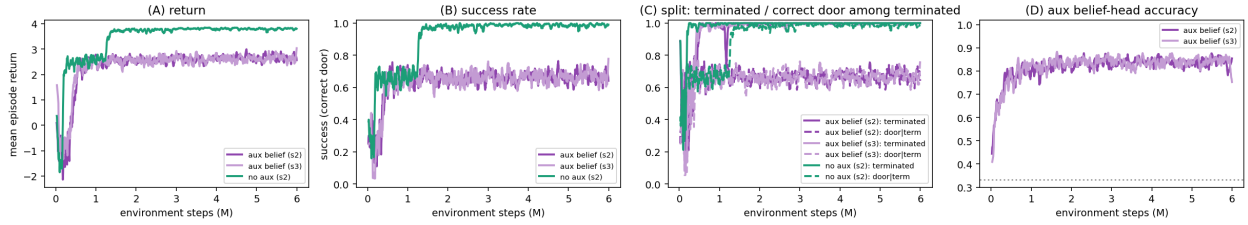


Figure 3: Training curves (aux-belief seeds 2 and 3, no-aux seed 2). (A) mean episode return; (B) success = category-correct door; (C) the success *split*: solid = fraction of episodes that terminate at *some* door (task completion, no timeout); dashed = correct-door rate *among* terminated episodes (decision quality); (D) auxiliary belief-head accuracy (aux runs only; chance 1/3).

The split in panel (C) carries the chapter’s two substantive training observations. First, *completion and decision quality are different skills with different learning curves*: both agents drive the terminated fraction (solid) to 1.0 within the first quarter of training — crossing the map, with its terrain-dependent skills, is learned quickly — while the correct-door rate among terminated episodes (dashed) develops later and more slowly. All remaining error is decision error, not failure to cross.

Second, and contrary to the motivating intuition, *the auxiliary belief loss did not help — it hurt*. The no-aux agent reaches ≈ 0.99 door accuracy; the aux agent plateaus at ≈ 0.69 , and a replicate aux run with a different seed lands at the same value (0.70). Inspecting the aux agents’ behaviour reveals what the plateau is: **both aux agents take the bottom door on 100% of episodes, in every category** — a constant door policy, which scores exactly the always-bottom baseline of 2/3 (balanced and lakes correct, rocky wrong). Meanwhile their auxiliary heads read the category at 0.77–0.85 accuracy throughout (panel D). The aux agents therefore *represent* the belief but never couple it to the door decision; the no-aux agent, which can only earn the last third of reward by conditioning on the category, escapes the constant-door local optimum and uses its belief. Representation supervision is no guarantee of behavioural integration — an early, purely behavioural instance of the representation-vs-use dissociation that the later chapters establish causally. Final performance:

	success	terminated	door-correct terminated
aux belief (seed 2)	0.69	1.00	0.69
aux belief (seed 3)	0.70	1.00	0.70
no aux (seed 2)	0.99	1.00	0.99

Table 1: End-of-training metrics (mean of last 10 logged iterations).

4 Behaviour: rollouts, generalisation, regimes

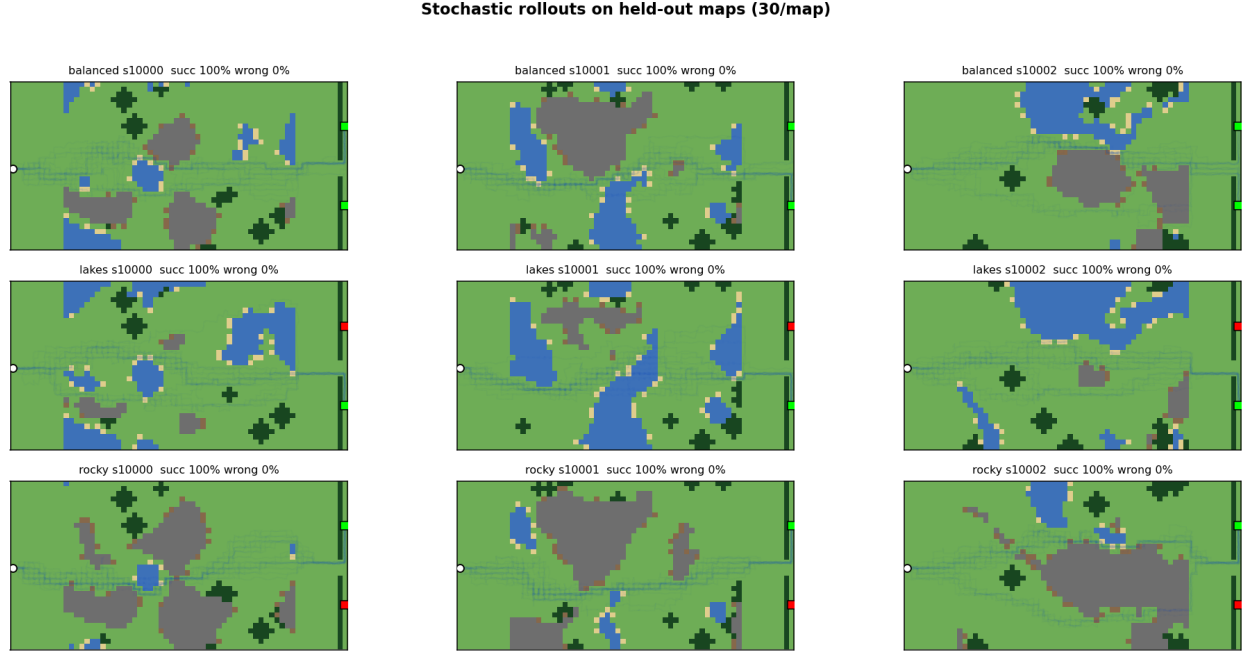


Figure 4: Stochastic rollouts on held-out maps (30 sampled-policy episodes per map; paths overlaid). Titles give success (correct door) and wrong-door rates.

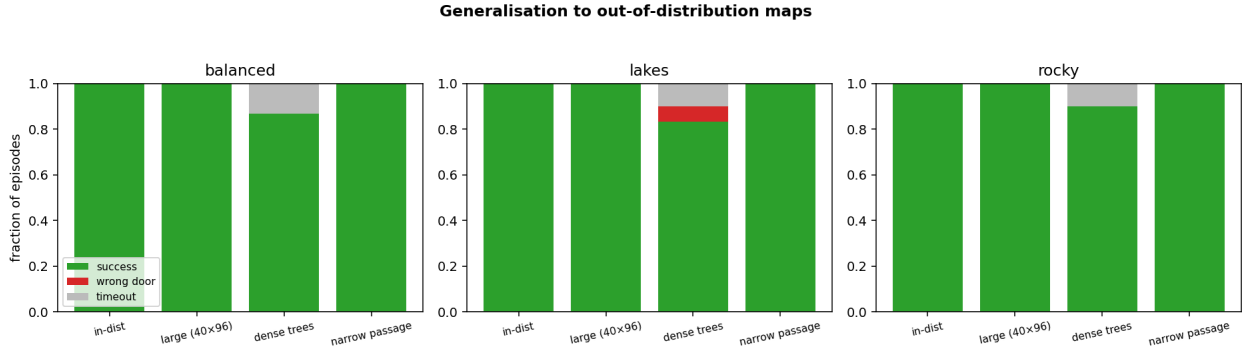


Figure 5: Out-of-distribution generalisation. Success / wrong-door / timeout on maps drawn outside the training distribution: larger maps (40×96 vs. 32×64), quadrupled tree density, and a single-cell wall passage.

Agent observations by behavioural regime
(egocentric view at the step the regime is expressed; agent = white dot)

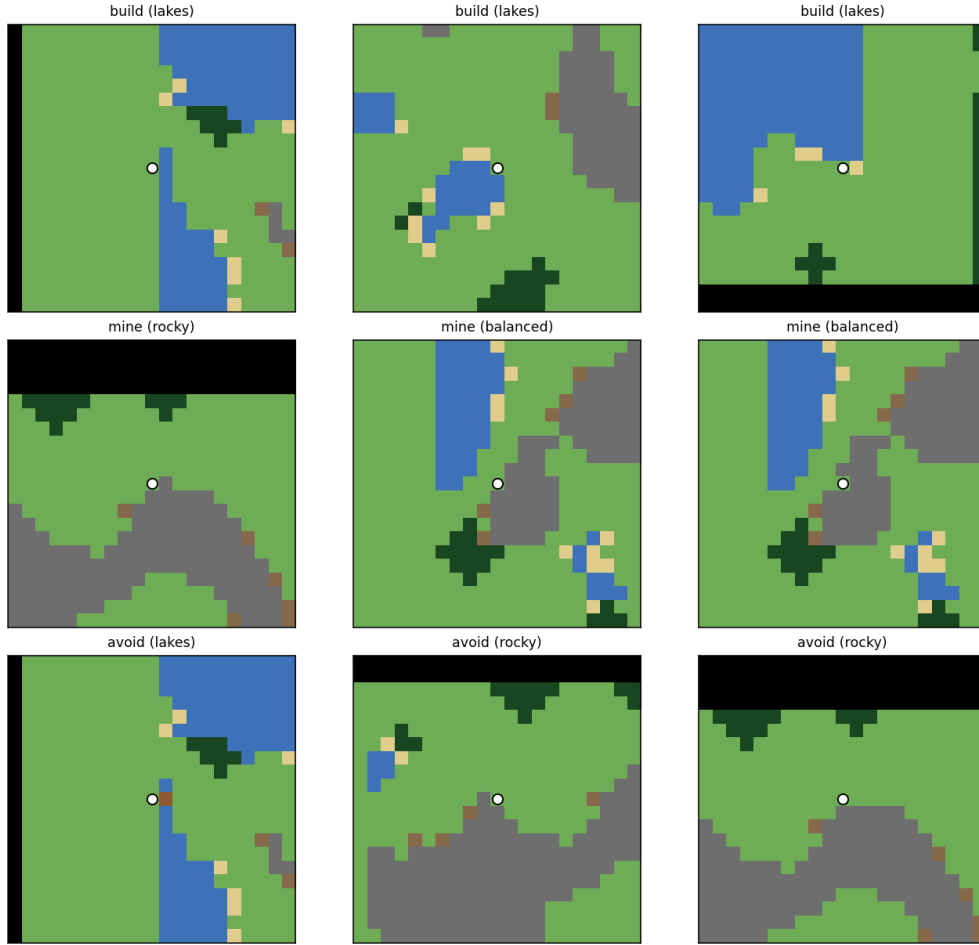


Figure 6: The three crossing regimes, from the agent’s eyes. Egocentric observations at the step each regime is expressed: *build* (facing water, placing a bridge tile), *mine* (facing rock, tunnelling), *avoid* (obstacle ahead, agent detours without using a skill).

Three points. First, behaviour is category-appropriate: on **lakes** maps the agent bridges, on **rocky** maps it tunnels, and on all maps it opportunistically detours around small obstacles (the *avoid* regime) — a substantial fraction of successful episodes use no skill at all. Second, generalisation is perfect along the size (40×96) and passage-width axes and degrades only under quadrupled tree density (0.83–0.90). Third, the OOD errors are informative about *which* competence breaks: under dense trees the failures are almost entirely *timeouts*, not wrong doors — the navigation problem gets harder, but the belief and its door read-out remain intact.

5 What the recurrent state encodes

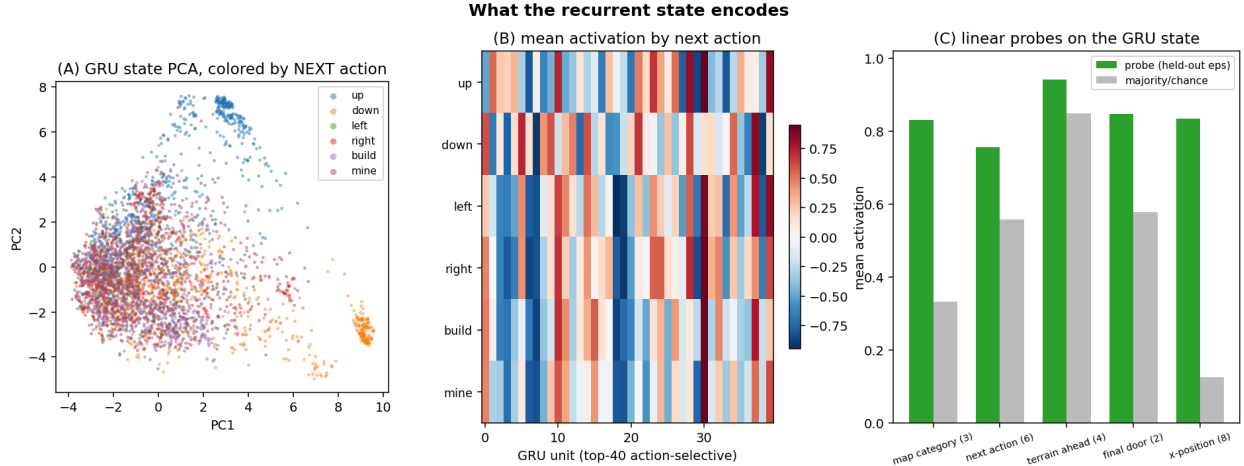


Figure 7: Hidden-state analysis (no-aux agent). (A) PCA of GRU states coloured by the *next* action — action selection is organised in a few directions of state space. (B) Mean activation of the 40 most action-selective units by next action: **build** and **mine** recruit distinct unit populations. (C) Linear probes (logistic regression, trained on half the episodes, tested on held-out episodes) against majority/chance baselines.

The probe battery makes the division of labour explicit: from the no-aux agent’s GRU state a linear read-out recovers the map category at 0.83 (chance 0.33), its position along the corridor at 0.83 (chance 0.125), the terrain ahead at 0.94, the next action at 0.76, and the door it will eventually choose at 0.85 — long before the doors are visible. Crucially, the category is decodable from an agent that was *never supervised on it*: the belief emerges from reward alone, because the door decision requires it.

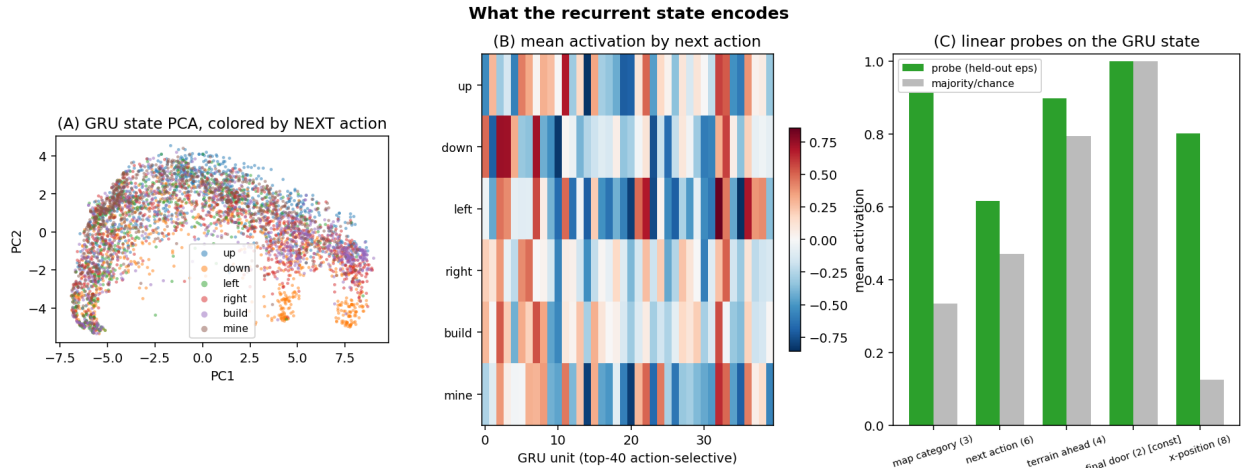


Figure 8: Same analysis for the aux-belief agent. The auxiliary loss yields an even sharper category code (0.92 vs. 0.83 probe accuracy) — yet this is the agent that takes the bottom door on every episode (§3); its “final door” probe is degenerate because the behaviour is constant. Decodability of the belief and behavioural use of the belief fully dissociate.

6 Belief along trajectories

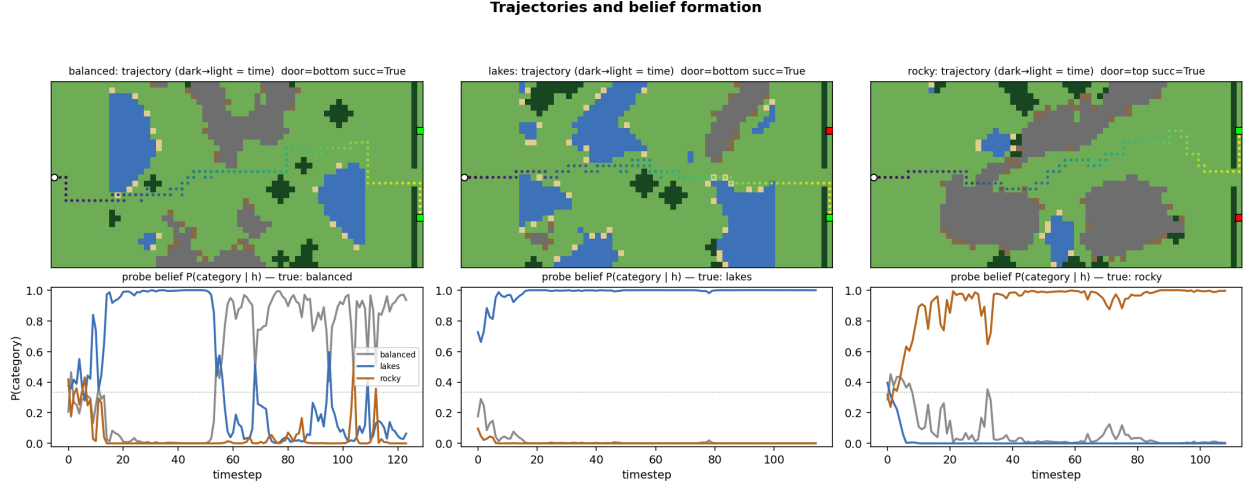


Figure 9: Trajectories and belief formation (no-aux agent). Top: single stochastic rollouts (colour = time). Bottom: the probe-decoded belief $P(\text{category} | h_t)$ over the same episodes. The belief is not written in one step: it accumulates as terrain evidence enters the view, can dip when the local view is ambiguous (a grassy stretch on a *lakes* map), and is held across the evidence-free corridor before the wall.

7 Does the agent know its plan?

Finally we ask a forward-looking question: when the agent approaches an obstacle it could either cross with a skill or walk around, does its hidden state already contain the choice? We build *toy maps* — a flat grass corridor with a single water or rock band of controlled extent and gap — roll the stochastic policy repeatedly, label each successful episode by the strategy it eventually expressed (*skill* vs. *around*), and train a probe to predict that label from the hidden state at fixed distances *before* the obstacle.

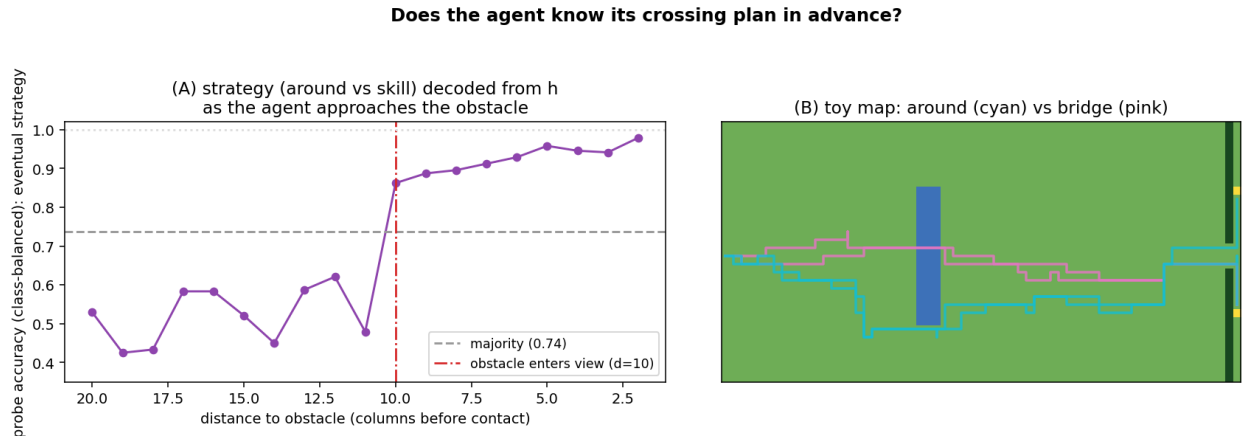


Figure 10: Strategy is decodable before the obstacle. (A) Held-out probe accuracy for the eventual strategy as a function of distance to the obstacle (right = closer). (B) Example toy map with *around* (cyan) and *bridge* (pink) episodes from identical starts.

The curve has a sharp, interpretable structure. While the obstacle is beyond the agent’s 21×21

view (distance > 10), accuracy sits at chance — there is nothing to know, since the toy corridor contains no earlier evidence. At $d = 10$, *exactly when the obstacle enters the field of view*, accuracy jumps to ≈ 0.86 , and it climbs to ≈ 0.98 by contact: the strategy is largely determined on first sight, several seconds of game time before the crossing is executed, and is then refined rather than reversed. The agent does not walk to the obstacle and react; it commits on approach. We are careful about the inferential limit: decodability before the obstacle shows the state *contains* the plan, not that the readout is causally used; the causal analyses of the following chapters take up exactly that question.

8 Summary

The chapter established the full experimental loop on **fork.wall**: a task whose latent category must be inferred (§1–2); agents whose completion and decision quality develop on different timescales, with the surprising twist that auxiliary belief supervision *trapped* two independent runs in a belief-blind constant-door optimum while the unsupervised agent coupled its belief to the decision (§3–4); a recurrent state that linearly encodes belief, action, position and the eventual decision — with belief decodability and belief *use* fully dissociating between the two agents (§5–6); and evidence that the state contains the agent’s crossing plan from the moment an obstacle enters view, well before execution (§7). These are the ingredients — and the measurement tools — for the causal belief-steering experiments of the next chapters.